Fantasy Premier League Points Forecasting and Squad Optimization

Project status report

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1 Conclusion and status

The system - a per-position machine-learning ensemble feeding a MILP squad optimizer - beats its thesis-era LSTM predecessor by 29% in realized backtest points over an identical 31-gameweek window (1,966 vs 1,526; Section 5.1). The current state of play, most decision-relevant first:

1. **CatBoost (MAE loss) is the best single forecaster at every position** (MASE 0.513-0.849), displacing the NNLS-blended ensemble, which now trails its own best member - a weight-fitting overfit under investigation after a same-day fix to how blend weights are estimated (Section 5.3, Section 6).
2. **The fixture-difficulty and minutes-projection features remain an unresolved trade-off**: they improved forecast accuracy at every position yet reduced realized backtest points 1,966 → 1,880 (Section 5.4) - the project's clearest evidence that the actual-points backtest, not forecast accuracy, must be the gold-standard check.
3. **No simple or econometric technique tested beats the ML models** (eight baselines, all reported in Section 5.2) - future gains must come from architecture, not from adding model types.
4. A same-day code-and-concept review found and fixed four methodological errors (blend-weight leakage, two live-mode staleness bugs, an evaluation asymmetry); the absolute backtest numbers predate the leakage fix and will be re-baselined (Section 6).

Priorities, in order: re-baseline the backtest under the fixed weight scheme; resolve the fixture/minutes divergence (noise-check on a second window, probabilistic-upside captaincy); re-examine the ensemble-vs-CatBoost gap; then a per-player model-selection pilot.

2 Introduction

The system predicts FPL player points one to several gameweeks ahead and converts forecasts into weekly squad decisions (transfers, lineup, captaincy, chips) via a MILP following Kristiansen et al. (2018). The governing principle is honest empirical validation: every candidate technique is tested against the production system on a fixed backtest window, and negative results are reported rather than discarded.

3 Data

Per-gameweek player statistics come from Vaastav Anand's public mirror of the official FPL API (`vaastav/Fantasy-Premier-League`), spanning six seasons - 2020-21 through 2025-26, 162,981 player-gameweek rows. The two oldest seasons lack the Opta expected-goals family and `starts` (introduced 2022-23); these are treated as missing, which the tree-based models handle natively.

Gameweeks carry a single ascending index (`GW_global`): dataset season N occupies gameweeks $(N-1)\times 38+1$ through $N\times 38$, so the fixed validation window - 2024-25 season, gameweeks 1-31 - is GW153-183.

Exploratory analysis (`notebooks/eda.ipynb`, all six seasons) establishes the two facts the methodology is built on: `total_points` is heavily right-skewed and zero-inflated at every position (Shapiro-Wilk rejects normality, $p<0.001$ throughout), motivating a scale-free error metric and models free of Gaussian assumptions; and the distribution is stable across seasons, with one caveat - the average defender score-per-gameweek series fails an Augmented Dickey-Fuller stationarity test over the full six seasons ($p=0.41$; the other positions pass), plausibly reflecting the 2025-26 `defensive_contribution` scoring change.

4 Methodology

4.1 Feature engineering

Each player-gameweek row carries roughly 80 engineered features:

- **Form.** Rolling 3- and 5-gameweek means, previous-gameweek values, and season-to-date expanding means over 18 per-gameweek statistics (points, minutes, xG family, BPS, ICT, price, ownership). All shifted one gameweek relative to the target row, so no row's features contain its own outcome.
- **Fixture difficulty.** The official FPL Fixture Difficulty Rating (1-5) of that gameweek's opponent, plus the mean over this and the next two fixtures. Deliberately *not* shifted: fixture lists are published ahead of play, so these are known-ahead inputs, not leakage.
- **Minutes projection.** Rolling 5-game start rate and 60+-minute rate, shifted like form - a "will he actually play?" signal.

4.2 Forecasting models

Four independent models, one per position (GK/DEF/MID/FWD) - the statistics that predict a goalkeeper's points are largely disjoint from a forward's, and pooling risks one position's scale dominating the loss. Per position, every regressor in a fixed registry is trained on the same features: LightGBM, XGBoost, CatBoost (MAE loss), OLS, Ridge, ElasticNet, PLS, Random Forest, Extra Trees, k-NN, LinearSVR, and a sample-capped RBF SVR. Boosted models take raw features including missing values; linear/kernel/distance models get a zero-imputer and standardizer. Registry members that tests find unhelpful are retained - the blend simply assigns them near-zero weight, preserving the comparison for later revisiting.

Predictions are combined per position by non-negative least squares (weights $w\geq 0$ minimizing $\left\|\sum_i w_i\hat{y}_i - y\right\|$, normalized to sum to one). Weight fitting is separated by purpose: *evaluation* weights are fit on one half of the held-out test window and scored on the other half, so reported ensemble accuracy is a genuine holdout figure; *production and backtest* weights are fit on a 16-gameweek window strictly before whatever is subsequently predicted, so no weight ever sees a gameweek it will be used to forecast. Plain OLS is designated the *index* - the simple benchmark every technique must beat, as a passive market index is the bar an active strategy must clear.

A parallel probabilistic view - per-position LightGBM quantile regressors (p10/p50/p90, crossing repaired by row-wise sorting) - produces a prediction interval per player-gameweek, because two players with equal expected points are not equal decisions once the 2x captain multiplier rewards upside. It does not yet feed the optimizer.

4.3 Evaluation design

Three layers, in increasing decision-relevance:

1. **Static split.** Train on GW $\leq$ 152, evaluate on GW153-183 - the window the original LSTM was validated on, keeping every comparison on one fixed window.
2. **Walk-forward validation.** For each gameweek from a start point, train strictly on earlier data and predict that gameweek only - many genuinely out-of-sample evaluations rather than one.
3. **Actual-points backtest (gold standard).** Walk-forward predictions run through the MILP over GW153-183; resulting squads scored with realized points. An accuracy gain that does not survive contact with the optimizer is not an improvement (Section 5.4).

Point forecasts are scored with MAE and, primarily, MASE (Hyndman & Koehler, 2006) - MAE relative to the in-sample error of a naive last-gameweek forecast, the scale fit on training data only; MASE $<$ 1 beats the naive floor regardless of the target's scale. Probabilistic forecasts are scored with pinball loss and [p10, p90] coverage against its nominal 0.80.

4.4 Squad optimization

The MILP follows Kristiansen et al. (2018): 15-player squad (2/5/5/3) under budget, max 3 per club, 11-player lineup under formation constraints, captain/vice-captain, limited free transfers with a 4-point penalty per extra, and one-shot chip logic (two wildcards, free hit, bench boost, triple captain), solved as a rolling horizon with only the first gameweek's decision locked in. Venter & van Vuuren (2024) - whose formulation matches almost exactly - attribute their strong case study (top 4.08% of 8.24M managers) to lookahead and forecast quality, not optimizer sophistication; hence this project's focus on forecasting.

5 Results

5.1 Backtest lineage

All systems scored by realized points over 2024-25 GW1-31, identical optimizer:

System	Actual points
Old LSTM + MILP (thesis)	1526
LightGBM + MILP, 4-season history	1811
Ensemble + MILP, 4-season history	1900
Ensemble + MILP, 6-season history	**1966**
1. fixture/minutes features	1880[1]

Extending history from four to six seasons was an unambiguous win: MASE fell at every position (FWD below 1.0 for the first time) and realized points rose 1,900 $\rightarrow$ 1,966.

5.2 Forecasting-technique experiments

Eight simple and econometric baselines - rolling mean, naive drift, SES, Holt, Theta, Croston, pooled AR(1), per-player ARIMA(1,0,1), and an empirical-Bayes shrinkage of player means toward position means - were each tested per position. None beats the ML models anywhere; SES is the best simple method, Croston and EB-shrinkage the weakest (full tables in

[1]Unresolved regression despite improved forecast accuracy (Section 5.4). Both this and the 1,966 figure predate the blend-weight fix of Section 6 and are modestly inflated; their relative comparison is unaffected.

RESEARCH_LOG.md). Plain OLS, the index, beats every simple baseline; the ML models in turn beat OLS.

5.3 Expanded model registry

Six techniques were added in one round (LinearSVR, RBF SVR, XGBoost, CatBoost, PLS, EB-shrinkage) and evaluated on the standard static split:

Position	CatBoost	LinearSVR	12-member ensemble	previous ensemble
GK	**0.513**	0.513	0.534	0.588
DEF	**0.705**	0.713	0.747	0.799
MID	**0.731**	0.751	0.806	0.799
FWD	**0.849**	0.869	0.853	0.959

MASE, GW153-183 static split. CatBoost uses MAE loss; ensemble scored on a genuine holdout half-window.

Two findings. First, **CatBoost with MAE loss is the best single model at every position**, by a wide margin over everything preceding it - consistent with the hypothesis (raised by LinearSVR's strong showing) that MAE-aligned training objectives suit a zero-inflated target with outlier hauls better than squared error. PLS and XGBoost added nothing (retained as near-zero-weight members). Second, **the blended ensemble now trails standalone CatBoost everywhere** - with 12 collinear members, NNLS overfits its half-window weight fit (the MID blend failed to select CatBoost at all). Whether the stricter production weight-fitting scheme (Section 4.2) closes this gap is under evaluation.

5.4 Fixture/minutes features: better forecasts, worse squads

Adding fixture-difficulty and minutes-projection features improved MASE at every position (DEF most, consistent with clean-sheet dependence) - yet realized backtest points *fell* 1,966 → 1,880. Leading hypothesis: the features smooth the mean forecast toward safe, nailed players, and the MILP - blind to variance - loses the high-ceiling captaincy differentials that drive realized hauls. The features are committed but not declared a net win; resolving this (noise-check on a second window, probabilistic-upside captaincy) is a top open item. The probabilistic module's [p10, p90] coverage is 0.88-0.93 against nominal 0.80 - somewhat wide, usable for relative risk ranking; conformal calibration is a possible refinement.

5.5 Exploratory branches

Two isolated branches, neither merged: a Bayesian belief-state MDP manager after Matthews et al. (2012) - 1,083 points myopic / 847 Q-learning where the production system scored 1,900, expected given documented simplifications - and a per-player model-selection plan (feasibility capped by 35% of live players having no prior-season history; recommendation: single-position pilot).

6 Methodological limitations

An internal code-and-concept review (2026-07-04) found four errors; all four were fixed the same day. Recorded here because published numbers predate some fixes:

1. **Blend-weight leakage (fixed).** Backtest predictions formerly reused blend weights fit inside the backtest window itself. Weights are now always fit on a window strictly before whatever is predicted. The 1,966/1,880 figures predate the fix and are modestly inflated; their relative comparison stands (both leaked identically). Re-baselining is pending.

2. **Live-mode staleness (fixed).** The weekly driver formerly reused each player's last played row, so live fixture-difficulty described last week's fixture and rolling form excluded the most recent match; API-vs-dataset team-name mismatches also silently dropped some teams' players. Live predictions now use per-gameweek API fixture difficulties and a synthetic next-gameweek row whose features include everything played. Backtests were never affected.
3. **Index-check asymmetry (fixed).** The ensemble-vs-OLS verdict now compares both on the same held-out rows.
4. **Optimizer rule currency (open).** The MILP caps banked free transfers at 2 (now 5 in FPL) and assumes sales at market value (real FPL: purchase price plus half profit). Fine for historical comparison; wrong for live 2026-27 play. Deliberately deferred - optimizer work is deprioritized in favor of forecasting.

7 References

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